Background

- In general, in clinical trials, the data collected for a subject can be classified into two types
  - data prior to treatment start
  - data on or after treatment start
- Treatment start date is generally considered as reference start date and is considered as day 1 of the study
- Instead of talking in absolute date values, it will be easy for reviewers if the dates are represented in relative values to this 'day 1'
  - For example, it is easy on the mind when interpreting the text 'from day 4 after treatment start the subject's blood pressure got lowered' when compared to the text 'from 18Jul2010 the blood pressure got lowered for the subject who started treatment on 15Jul2010'.
- For this relative day values, all collections before reference start date are given a negative integer value starting from -1 (to -2, -3 and so on)
- And, all collections on or after reference start date are given positive integer value starting from 1
- There is no 'day 0' as per CDISC SDTM standard
- SDTM standard allows for creation of the 'study day' variables to capture the relative day of the observation starting with the reference date as Day 1
- These study day variables end with 'DY' using the same fragment from the dependent date variable
  - Relative day of LBDTC is stored in LBDY variable
  - Relative day of AESTDTC is stored in AESTDY variable
  - Relative day of CMENDTC is stored in CMENDY variable

What do we learn in this lesson?

- In this lesson, we will see how to derive study day variables in SDTM datasets

Programming flow visualization

- We have our input dataset with some date values in it
USBJID	Date
- We have our reference start date in a second dataset
USBJID	RFSTDTC
- We need to pull the reference start date into our input dataset by merging
USBJID	Date	RFSTDTC
- We will now have our date and reference start date available side by side for comparison
USBJID	Date	RFSTDTC	Study day
			Date-RFSTDTC
			Date-RFSTDTC+1
- We will then derive our study variable
USBJID	Date	RFSTDTC	Study day
			-y
			-x
			a
			b
			c